



CryptoCopilot

An intelligent assistant for the world of crypto

ML direction signals · technical analysis · cited AI chat · safe paper trading

Demo Day · June 2026

Decision-support, not financial advice

A polyglot system: Python (data + ML) · Java/Spring Boot · React — five containers, one shared database.

Money, Control — and a Different Idea

Traditional money

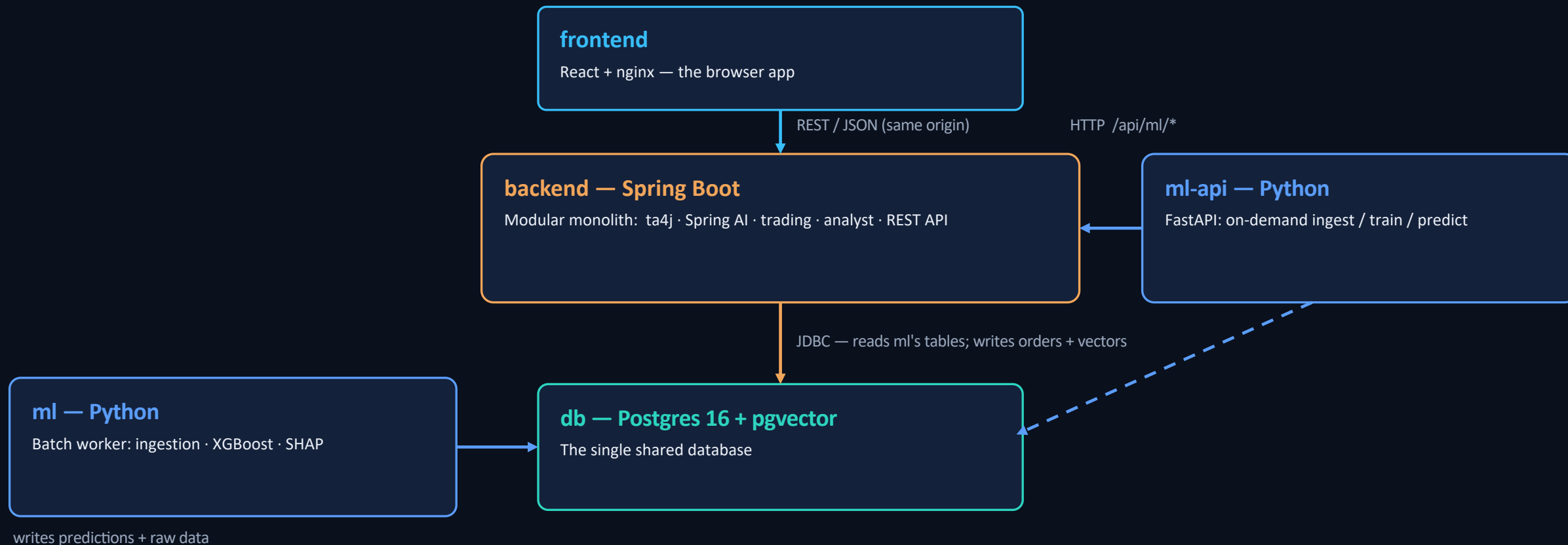
-  Controlled by central banks — central control.
-  You pay fees to hold and move your own money.
-  Someone else is always in charge of your account.

Cryptocurrency

-  Decentralized — no central authority in control.
-  You hold your own money, directly.
- + Pros: open, global, fewer gatekeepers.
- Cons: complex, volatile, confusing for newcomers.

CryptoCopilot helps newcomers: it turns scattered market data + news into clear information and signals about coins.

One System, Two Languages, Five Containers



The database is the polyglot boundary: Python writes → Java reads. No RPC, no shared model files.

Real Data: 10 Coins, 5 Free Sources

BTC · ETH · SOL · BNB · XRP · ADA · AVAX · DOT · LINK · MATIC

Where it comes from

-  Binance — price candles (1h / 4h / 1d, ~2 years)
-  CoinGecko — market cap, supply, community & dev data
-  RSS x5 — CoinDesk, Cointelegraph, Decrypt, The Block...
-  Blockchain.com + Etherscan — on-chain activity
-  Curated knowledge base — how each coin works

How we treat it

-  ~231,000 rows loaded into one Postgres database
-  Multi-source by design — no single point of failure
-  If any source fails: log and skip, never crash
-  One writer per table — clean, auditable data
-  All sources are public and free

An Honest Edge, Not a Crystal Ball



- ▶ Predicts UP / FLAT / DOWN over the next 24 hours (a $\pm 2\%$ move).
- ▶ ROC-AUC = 0.578 — a small but real edge above luck (0.5 = a coin flip).
- ▶ Calibrated probabilities: “70%” really means about 70%.
- ▶ SHAP explains the top-3 reasons behind every single prediction.
- ▶ Trained on ~2 years of history — the score is reported honestly.

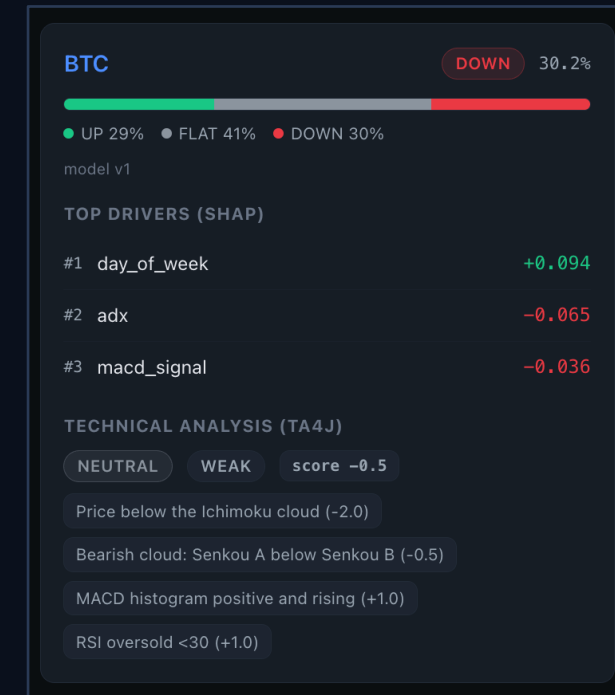
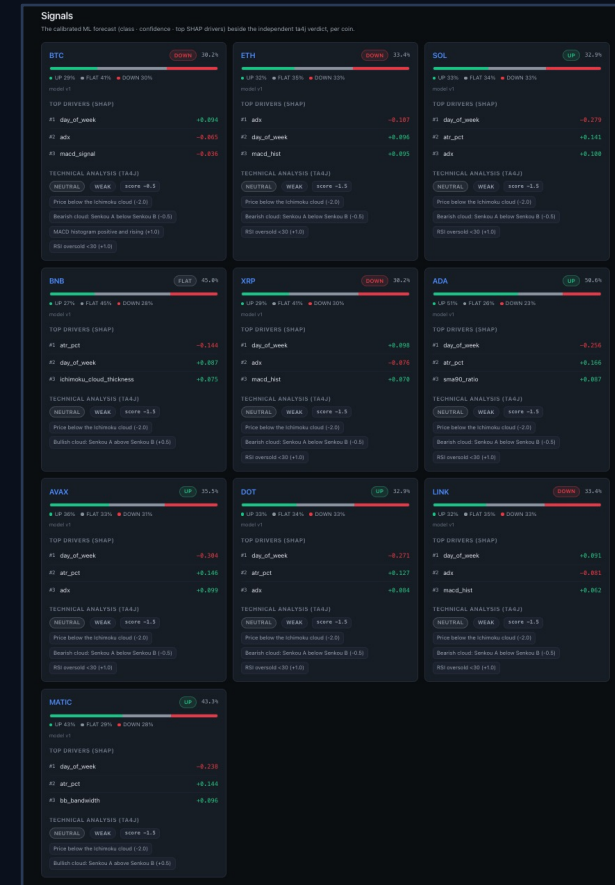


Chart Signals You Can Audit

- ▶ Computed in Java with ta4j, straight from the price candles.
- ▶ Indicators: Ichimoku · RSI · MACD · Bollinger · ATR.
- ▶ Deterministic rules → BULLISH / NEUTRAL / BEARISH + a confidence.
- ▶ It shows the exact rules that fired — fully transparent.
- ▶ Independent of the ML model — a real second opinion.

Backend = one modular monolith (Spring Boot), not microservices.



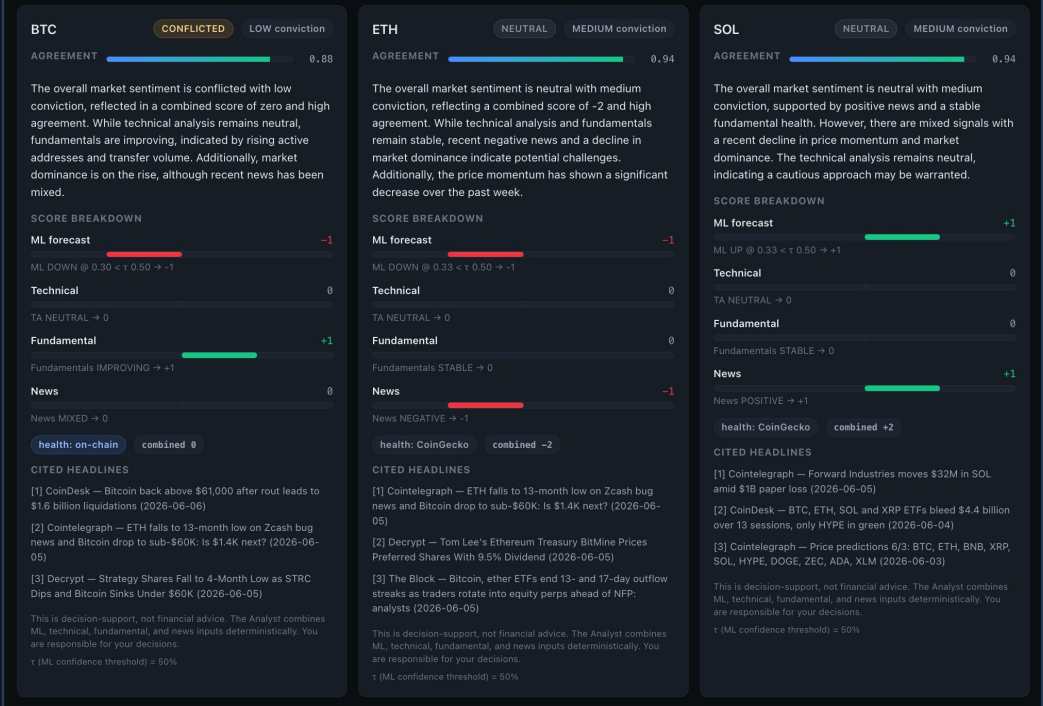
AI Chat That Cites Its Sources

- ▶ Embeddings turn text into numbers that capture meaning.
- ▶ pgvector stores them inside the same Postgres database.
- ▶ RAG = retrieve real sources, then answer using only them.
- ▶ Every answer is cited [1][2]; it refuses if the answer isn't there.
- ▶ Never gives trading advice — by design.
- ▶ $\text{recall@8} = 0.90 \cdot 100\%$ of answers cited.

Runs free & local (Ollama llama3.2:3b) — or switch to OpenAI gpt-4o-mini with one click.

Analyst

One fused, deterministic opinion per coin — ML + technical + fundamental + news — with a guarded summary. Click a card for the full fundamental snapshot.



Four Views → One Clear Opinion



Analyst → Direction · Conviction · Agreement

- ▶ A simple, deterministic score from -2...+2 per view.
- ▶ Same inputs always give the same verdict — auditable.
- ▶ A hallucination guard keeps the summary truthful.
- ▶ Works even with the AI switched off (plain template).

Analyst

One fused, deterministic opinion per coin — ML + technical + fundamental + news — with a guarded summary. Click a card for the full fundamental snapshot.

BTC

CONFLICTED

LOW conviction

AGREEMENT

0.88

The overall market sentiment is conflicted with low conviction, reflected in a combined score of zero and high agreement. While technical analysis remains neutral, fundamentals are improving, indicated by rising active addresses and transfer volume. Additionally, market dominance is on the rise, although recent news has been mixed.

SCORE BREAKDOWN

ML forecast

-1

ML DOWN @ 0.30 < τ 0.50 → -1

Technical

0

TA NEUTRAL → 0

Fundamental

+1

Fundamentals IMPROVING → +1

News

0

News MIXED → 0

health: on-chain

combined 0

CITED HEADLINES

[1] CoinDesk — Bitcoin back above \$61,000 after rout leads to \$1.6 billion liquidations (2026-06-06)

[2] Cointelegraph — ETH falls to 13-month low on Zcash bug news and Bitcoin drop to sub-\$60K: Is \$1.4K next? (2026-06-05)

[3] Decrypt — Strategy Shares Fall to 4-Month Low as STRC Dips and Bitcoin Sinks Under \$60K (2026-06-05)

This is decision-support, not financial advice. The Analyst combines ML, technical, fundamental, and news inputs deterministically. You are responsible for your decisions.

τ (ML confidence threshold) = 50%

ETH

NEUTRAL

MEDIUM conviction

AGREEMENT

0.94

The overall market sentiment is neutral with medium conviction, reflecting a combined score of -2 and high agreement. While technical analysis and fundamentals remain stable, recent negative news and a decline in market dominance indicate potential challenges. Additionally, the price momentum has shown a significant decrease over the past week.

SCORE BREAKDOWN

ML forecast

-1

ML DOWN @ 0.33 < τ 0.50 → -1

Technical

0

TA NEUTRAL → 0

Fundamental

0

Fundamentals STABLE → 0

News

-1

News NEGATIVE → -1

health: CoinGecko

combined -2

CITED HEADLINES

[1] Cointelegraph — ETH falls to 13-month low on Zcash bug news and Bitcoin drop to sub-\$60K: Is \$1.4K next? (2026-06-05)

[2] Decrypt — Tom Lee's Ethereum Treasury BitMine Prices Preferred Shares With 9.5% Dividend (2026-06-05)

[3] The Block — Bitcoin, ether ETFs end 13- and 17-day outflow streaks as traders rotate into equity perps ahead of NFP: analysts (2026-06-05)

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τ (ML confidence threshold) = 50%

SOL

NEUTRAL

MEDIUM conviction

AGREEMENT

0.94

The overall market sentiment is neutral with medium conviction, supported by positive news and a stable fundamental health. However, there are mixed signals with a recent decline in price momentum and market dominance. The technical analysis remains neutral, indicating a cautious approach may be warranted.

SCORE BREAKDOWN

ML forecast

+1

ML UP @ 0.33 < τ 0.50 → +1

Technical

0

TA NEUTRAL → 0

Fundamental

0

Fundamentals STABLE → 0

News

+1

News POSITIVE → +1

health: CoinGecko

combined +2

CITED HEADLINES

[1] Cointelegraph — Forward Industries moves \$32M in SOL amid \$1B paper loss (2026-06-05)

[2] CoinDesk — BTC, ETH, SOL and XRP ETFs bleed \$4.4 billion over 13 sessions, only HYPE in green (2026-06-04)

[3] Cointelegraph — Price predictions 6/3: BTC, ETH, BNB, XRP, SOL, HYPE, DOGE, ZEC, ADA, XLM (2026-06-03)

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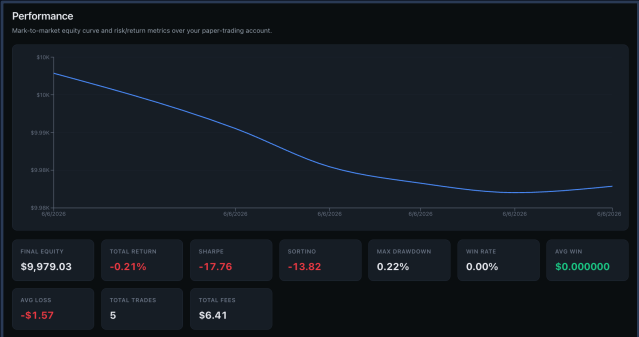
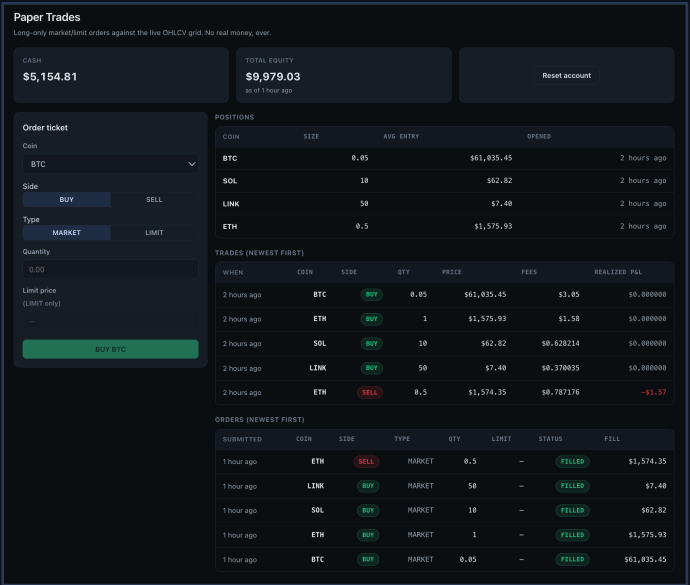
τ (ML confidence threshold) = 50%

Decision-support, not financial advice · Paper trading only

8 / 12

Practice Trading, No Real Money

- ▶ Paper trading = real mechanics, fake money.
- ▶ Long-only; realistic fills with slippage + a 0.1% fee.
- ▶ Market & limit orders; positions, trades, account tracked.
- ▶ Performance: equity curve + Sharpe, Sortino, drawdown.
- ▶ Honest backtest result — the engine is correct, not magic.



A Clean, Typed Web App

- ▶ React · Vite · TypeScript · nginx.
- ▶ Candlestick charts with klinecharts; equity with Recharts.
- ▶ 7 pages: Markets, Signals, Analyst, Chat, Trades, Performance...
- ▶ ...plus an ML Pipeline page — run the whole pipeline in-browser.
- ▶ A disclaimer is on every page.

Decision-support, not financial advice. Paper trading only — no real money, ever.

CryptoCopilot
Decision support

Markets

The 10-coin universe — price and 24h change from 4h candles, latest market cap from CoinGecko.

ML Pipeline

COIN	PRICE	24h	MARKET CAP
BTC	\$58,898.81	-1.73%	\$1.2T
ETH	\$1,578.48	-5.59%	\$182.6B
SOL	\$62.59	-5.28%	\$36.31B
BNB	\$578.22	-2.81%	\$77.94B
XRP	\$1.89	-2.58%	\$67.68B
ADA	\$0.156788	-3.51%	\$5.83B
AVAX	\$6.64	-6.16%	\$2.69B
DOT	\$8.941888	-5.33%	\$1.59B
LINK	\$7.37	-2.45%	\$5.37B
MATIC	\$0.875888	-2.74%	\$818.86M

Answering model: [OpenAI](#)
Options: [OpenAI](#)

Paper trading only. No real money, ever.

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CryptoCopilot
Decision support

INGEST DATA

Crawl the five public sources (Binance OHLCV, CoinGecko, RSS news, on-chain, fundamentals) into Postgres. Takes a few minutes.

[Run ingestion](#) Last ingested: 1 hour ago

TRAIN MODEL

Build features, tune + fit the calibrated XGBoost, evaluate on the held-out test split, and save model(u/v). Takes a few minutes.

[Train model](#) Last trained: 1 hour ago

PREDICT

Write the latest 24h direction forecast + top-3 SHAP drivers for all 10 coins to the predictions table.

[Run predict](#) Last predicted: 1 hour ago

LATEST PREDICTIONS

COIN	CLASS	P(UP)	P(FLAT)	P(DOWN)	AS OF
ADA	UP	58.6%	26.3%	22.9%	2 hours ago
AVAX	UP	35.5%	33.4%	31.5%	2 hours ago
BNB	FLAT	26.7%	45.8%	28.3%	2 hours ago
BTC	DOWN	28.5%	41.3%	38.2%	2 hours ago
DOT	UP	32.9%	34.4%	32.8%	2 hours ago
ETH	DOWN	31.5%	35.8%	33.4%	2 hours ago
LINK	DOWN	31.5%	35.8%	33.4%	2 hours ago
MATIC	UP	43.3%	29.8%	27.7%	2 hours ago
SOL	UP	32.9%	34.4%	32.8%	2 hours ago
XRP	DOWN	28.5%	41.3%	38.2%	2 hours ago

Answering model: [OpenAI](#)
Options: [OpenAI](#)

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What It Achieves — Reported Honestly

Machine learning

ROC-AUC 0.578

A small, real edge above luck.

AUC + Brier pass; F1 limited by ~2y data.

RAG retrieval

recall@8 0.90

100% of answers cited.

Refuses the unknown; ≈ €0, local.

Engineering

5 containers · 1 DB

Tests + CI across 3 languages.

Backtest honest, engine correct.

The goal was production-grade polyglot engineering — built end-to-end and reported honestly.

- ▶ Hard rules, never broken: paper only · crypto only · no shorts/leverage · decision-support, not advice.

Thank you!

Questions?

CryptoCopilot — four views on every coin, fused into one explainable opinion.

app · localhost:3000

API docs · :8080/swagger-ui.html

ML API · :8000/docs

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